LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 1 Test (Set B): Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

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SECTION A

Q1. Which of the following is NOT a chemical change? **[1]**

Answer: (d) Melting of ice (only the state changes; no new substance is formed)

Q2. In the reaction $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, the substance that is oxidised is: **[1]**

Answer: (a) H_2S. It loses hydrogen.

Q3. Which of these is an endothermic process? **[1]**

Answer: (b) Decomposition of calcium carbonate (it needs heat)

Q4. When lead nitrate is heated strongly, the products are: **[1]**

Answer: (a) lead oxide, nitrogen dioxide and oxygen. $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

Q5. A balanced chemical equation has equal numbers of: **[1]**

Answer: (b) atoms of each element on both sides

Q6. **Assertion (A):** Packets of chips are flushed with nitrogen gas. **Reason (R):** Nitrogen is a highly reactive gas that kills bacteria. **[1]**

Answer: (c). Nitrogen is used because it is unreactive. It replaces oxygen and prevents rancidity.

SECTION B

Q7. What is an oxidation reaction? Identify the substance oxidised in: $4Na + O_2 \rightarrow 2Na_2O$. **[2]**

Answer: A reaction in which a substance gains oxygen (or loses hydrogen). In $4Na + O_2 \rightarrow 2Na_2O$, sodium is oxidised, because it gains oxygen.

- Definition: 1
- Sodium oxidised, with reason: 1

OR, answer: (a) $CaCO_3 \rightarrow CaO + CO_2$ (on heating). (b) $2AgBr \rightarrow 2Ag + Br_2$ (in sunlight).

- Each correct equation: 1

Q8. Why do we paint iron articles? Name the process that this prevents. **[2]**

Answer: Paint forms a layer that stops air (oxygen) and moisture from reaching the iron. This prevents rusting, which is a type of corrosion.

- Blocks air and moisture: 1
- Prevents rusting/corrosion: 1

SECTION C

Q9. A shiny brown element X becomes black when heated in air. **[3]**

(i) X is copper (Cu). The black substance is copper(II) oxide (CuO). [1]
(ii) $2Cu + O_2 \rightarrow 2CuO$ [1]
(iii) Combination reaction (also an oxidation of copper). [1]

Q10. Explain each of the following with one balanced equation: (i) displacement reaction, (ii) double displacement reaction, (iii) redox reaction. **[3]**

Answer: (i) A more reactive element displaces a less reactive one from its compound: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. (ii) Two compounds exchange ions: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$. (iii) One substance is oxidised while another is reduced: $ZnO + C \rightarrow Zn + CO$ (C is oxidised, ZnO is reduced).

- Each type with equation: 1

OR, answer: (a) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (b) $CaO + H_2O \rightarrow Ca(OH)_2$ (c) $2H_2O \rightarrow 2H_2 + O_2$

- Each balanced equation: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Corrosion is the slow eating away of a metal when it reacts with substances around it, such as air, moisture and acids. Rust is hydrated iron(III) oxide ($Fe_2O_3 \cdot xH_2O$). Iron rusts only when both air (oxygen) and water are present. Prevention (any two): painting, oiling or greasing, galvanising, making alloys. [3]

- Definition: 1
- Rust + conditions: 1
- Two methods: 1

(ii) Fats and oils in food get oxidised, so the food's smell and taste change. Prevention (any two): add antioxidants; store in airtight containers; flush packets with nitrogen; keep in a refrigerator. [2]

- Definition: 1
- Two ways: 1

OR, answer: (a) (i) $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$ (ii) $BaCl_2 + K_2SO_4 \rightarrow BaSO_4 + 2KCl$ (iii) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (already balanced). (b) Mass can neither be created nor destroyed in a chemical reaction (law of conservation of mass), so the number of atoms of each element must be the same on both sides.

- (a) 1 each
- (b) law of conservation of mass 1, equal atoms 1

SECTION E

Q12. A student filled a plastic mug with water and added a few drops of dilute sulphuric acid. Two carbon electrodes were fixed in the mug and connected… **[4]**

(i) Pure water is a very poor conductor of electricity. The acid provides ions so that current can flow. [1]

(ii) Hydrogen (at the cathode). [1]

(iii) $2H_2O \rightarrow 2H_2 + O_2$. This is a decomposition (electrolytic) reaction. The smaller volume is oxygen, which relights a glowing splinter. [2]

- Equation + type: 1
- Oxygen test: 1